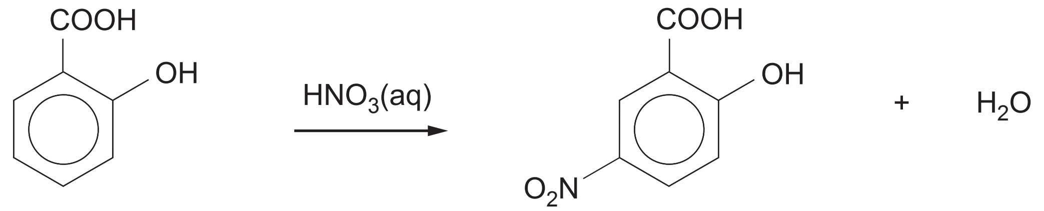
Examiner only

**11.** *(a)* 2-Hydroxy-5-nitrobenzenecarboxylic acid is made by reacting 2-hydroxybenzenecarboxylic acid and aqueous nitric acid.




(i) A basic outline of the method is given below.

Add 4.00 g of 2-hydroxybenzenecarboxylic acid to 100 $cm^3$ of nitric acid of concentration 2 mol $dm^{-3}$. Heat the mixture to 60 °C for 10 minutes. As the reaction proceeds brown fumes containing toxic nitrogen dioxide are produced. Add the products to 300 $cm^3$ of cold water. A yellow solution and white crystals of 2-hydroxy-5-nitrobenzenecarboxylic acid are obtained.

Use this outline method to produce a more detailed method suitable for others to use to be able to produce dry white crystals of the acid. As part of your answer you should give the type and size of apparatus used and mention any necessary health and safety factors.

Details of subsequent recrystallisation are **not** required. [6 QER]

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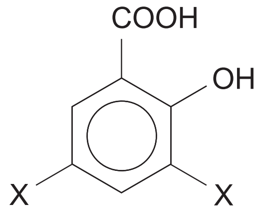
(ii) In this experiment the percentage yield of dry 2-hydroxy-5-nitrobenzenecarboxylic acid ($M_r$ 183) was 41 %.

Calculate the mass produced based on the original mass of the 2-hydroxybenzenecarboxylic acid ($M_r$ 138). [3]

Mass = .......................................................... g

(iii) Several other products were obtained during this reaction. One of these products is compound **J**.

Spectroscopy and analysis show that compound **J** is a substituted hydroxybenzenecarboxylic acid of $M_r$ 228. Each molecule contains seven oxygen atoms.




compound **J**

Use the information given to suggest a structure for compound **J**. Show your reasoning. [2]

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(iv) The yield of 2-hydroxy-5-nitrobenzenecarboxylic acid in this experiment is low.

Suggest how the experiment could be modified to increase the yield of the required product. [1]

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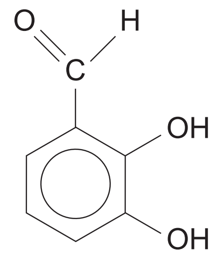
*(b)* 2,3-Dihydroxybenzaldehyde has the same molecular formula as 2-hydroxybenzenecarboxylic acid.




Suggest a simple reaction that will give a positive result for 2-hydroxybenzenecarboxylic acid but **not** for 2,3-dihydroxybenzaldehyde. You should give the reagent(s) used and the result of the test with the acid. [1]

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